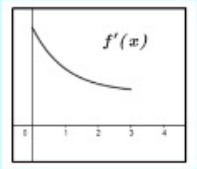
FUN-6.B.1 An antiderivative of a function f is a function g whose derivative is f.
FUN-6.B.2 If a function f is continuous on an interval containing a, the function defined by $F(x) = \int_a^x f(t)dt$ is an antiderivative of f for x in the interval.
FUN-6.B.3 If f is continuous on the interval $[a, b]$ and F is an antiderivative of f, then $\int_a^b f(x)dx = F(b) - F(a)$

6-7 The Fundamental Theorem of Calculus and Definite Integrals

Consider the graph of $f'(x)$ shown in the figure on the right.



What does the expression $\int_0^2 f'(x)dx$ represent?

The area bounded by the graph of $f'(x)$ and the $x-$axis on the interval [0,2].

We have examined area under a curve in many ways so far. We've looked at contextual problems to understand what this area represents in specific situations. We've also discussed multiple ways to find or approximate the area.

However, we want to find the exact areas under curves even if the area is NOT made of geometric shapes. To do this, we will use the **Fundamental Theorem of Calculus (part 1)**.

Fundamental Theorem of Calculus (part 1)
If the function $f'(x)$ is continuous on the closed interval $[a, b]$, then $\int_a^b f'(x)dx = f(b) - f(a)$
If the function $f(x)$ is continuous on the closed interval $[a, b]$, then $\int_a^b f(x)dx = F(b) - F(a)$

Notation is very important. Your answer must be the **antiderivative of the function in the integral.

Definition of an antiderivative	An **antiderivative** of a function $f(x)$ is a function $F(x)$ whose derivative is $f(x)$.

$F'(x) = f(x)$

$F(x)$

derivative $f(x)$

$f'(x)$

$f''(x)$

antiderivative

Example 1: Use the Fundamental Theorem of Calculus to determine an expression for each integral:

a.) $\int_1^5 g'(x)\,dx = g(5) - g(1)$

b.) $\int_5^{-2} h''(x)\,dx = h'(-2) - h'(5)$

you don't have to flip
- you don't need the smaller number as the lower limit

What are the **derivatives** of the following functions:

$\frac{d}{dx}(8x) =$ 8

$\frac{d}{dx}(-5x) =$ -5

$\frac{d}{dx}(1.3x) =$ 1.3

$\frac{d}{dx}\left(\frac{2}{3}x\right) =$ $\frac{2}{3}$

What are the **antiderivatives** of the following functions:

$f(x) = 7$ $\quad F(x) =$ $7x$

$f(x) = -8$ $\quad F(x) =$ $-8x$

$f(x) = \pi$ $\quad F(x) =$ πx

This leads us to our first rule of antidifferentiation:

Antiderivative (or integral) of a constant	If c is a constant, then $\int c \ dx = cx$

Example 2: The function $f(x)$ and its derivatives are continuous for all real numbers. Selected values of F, f, f', and f'' are given in the table below. Use the table to answer the following:

x	1	3	6	7	13
$F(x)$	3	6	-2	5	-7
$f(x)$	2	7	-3	4	10
$f'(x)$	8	0	-1	-3	4
$f''(x)$	4	1	7	-5	-11

a.) $\int_1^6 f'(x)\,dx = f(6) - f(1) = -3 - 2 = \boxed{-5}$

Example 2: The function $f(x)$ and its derivatives are continuous for all real numbers. Selected values of F, f, f', and f'' are given in the table below. Use the table to answer the following:

x	1	3	6	7	13
$F(x)$	3	6	-2	5	-7
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$f'(x)$	8	0	-1	-3	4
$f''(x)$	4	1	7	-5	-11

b.) $\int_7^1 f''(x)\,dx = f'(1) - f'(7) = 8 - -3 = 11$

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c.) $\int_6^3 -2f(x)\,dx = -2\int_6^3 f(x)dx$

$= -2[F(3) - F(6)]$

$= -2(6 - -2)$

$= -16$

Example 2: The function $f(x)$ and its derivatives are continuous for all real numbers. Selected values of F, f, f', and f'' are given in the table below. Use the table to answer the following:

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$F(x)$	3	6	-2	5	-7
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$f''(x)$	4	1	7	-5	-11

d.) $\int_6^7 [f(x) - 2]\, dx = \left[F(x) - 2x\right]\Big|_6^7$

$= F(7) - 2\cdot 7 - (F(6) - 2\cdot 6)$

$= 5 - 14 - (-2 - 12)$ ←

$= -9 + (+14)$

$= 5$

Example 2: The function $f(x)$ and its derivatives are continuous for all real numbers. Selected values of F, f, f', and f'' are given in the table below. Use the table to answer the following:

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$F(x)$	3	6	-2	5	-7
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e.) $\int_3^{13} [f'(x) + 3]\, dx = \left[f(x) + 3x\right]\Big|_3^{13}$

$= f(13) + 39 - (f(3) + 9)$

$= 10 + 39 - (7 + 9)$

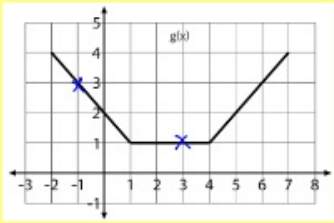
$= 33$

Example 3: The graph of $g(x)$ consists of three line segments and is given below. Use the graph of $g(x)$ to find the following:

a. $\int_{-1}^{3} g'(x)dx = g(3) - g(-1)$

$= 1 - 3$

$= -2$

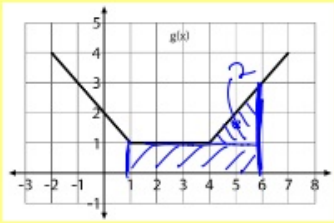


Example 3: The graph of $g(x)$ consists of three line segments and is given below. Use the graph of $g(x)$ to find the following:

b. $\int_{6}^{1} g(x)dx = -\int_{1}^{6} g(x)dx$

$= -[5+2]$

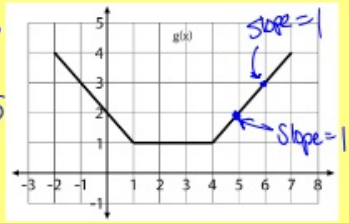
$= -7$



Example 3: The graph of $g(x)$ consists of three line segments and is given below. Use the graph of $g(x)$ to find the following:

c. $\int_5^6 [g''(x) + 4]dx = \left[g'(x) + 4x\right]\Big|_5^6$

$= g'(6) + 24 - (g'(5) + 20)$

$= 1 + 24 - (1 + 20)$

$= 4$